

# Revised jamovi cheat sheet

This cheat sheet outlines key jamovi commands and navigation steps for statistical analysis, as taught in PHCM9795 Foundations of Biostatistics.

## Generative AI statement

This work was developed with the assistance of ChatGPT (OpenAI).

### 1. jamovi Interface & Setup

- **Download & Install:** Free at <https://www.jamovi.org/download.html>. Version 2.7.24 current is recommended. [Section 1.9]
- **Main Views:** **Spreadsheet view** (accessed via **Data** tab) for data entry and changes; **Analysis view** (accessed via **Analyses** tab) for statistical results. [Section 1.10]
- **Open/Save/Export:** Click the **three-bar icon** (upper-left corner). [Section 1.10]
  - **Open Data:** 3-bar icon > **Open** > **Browse**.
  - **Save Work:** 3-bar icon > **Save** (.omv file format, saves data and output).
  - **Export Data:** 3-bar icon > **Export** (data in other formats like **txt**, **csv**, **xlsx**, **rds**, **dta**).
  - **Export Output:** 3-bar icon > **Export** (output as **pdf** or **html**).
  - **Copy Output to Word Processor:** **Right-click** on output and choose **Export** (plain text) or **Copy table/Copy table as HTML**.
- **Install Modules:** **Analyses tab** > **large + (top-right)** > **jamovi library**. Search and **INSTALL** desired module (e.g., **survey**, **esci**). [Sections 2.15 and 3.18]

### 2. Data Handling & Manipulation

- **Enter Data:** Type directly into cells in the spreadsheet view. [Section 1.10]
- **Rename Variables:** Click **Data tab** > **click column header** > **Setup**, then edit the **Variable name**, **Measure type** (e.g., Nominal, Continuous), and **Levels** (e.g., for categorical data). [Section 1.10]
- **Create New Variable:** **Data tab** > **click empty column** > **Setup** > **NEW COMPUTED VARIABLE**. Enter variable name and formula in the **fx** box (e.g., **weight / height^2**). [Section 3.14]
- **Set Value to Missing:** Click the cell with the erroneous value and press **Delete** or **Backspace**. Always keep an original, unedited copy of your data. [Section 1.12]
- **Weight Data** (for Summarised Categorical Data): **Data tab** > **Weights**, then define the count variable as the **weight variable**. [Section 6.10]
- **Re-order Categorical Variable Levels:** In **Data** > **Setup**, select the variable, then in the **Levels** section, click and drag levels or use the arrows to re-order (e.g., to set “Active” as the first level for a group variable). This is crucial for correct calculation of relative risks and odds ratios. [Section 6.8]

### 3. Descriptive Statistics (Numerical)

- **Continuous data:**
  - **Analyses > Exploration > Descriptives.** Move variable(s) to **Variables** box. [Section 1.11]
  - **Interquartile Range (IQR):** In **Statistics** section, tick **Percentiles**. [Section 1.11]
  - **Standard Error of Mean and Confidence Interval for Mean:** In **Statistics** section, tick **Std. error of Mean** and **Confidence interval for Mean**. [Section 3.17]
  - **Summarise by Another Variable:** Move grouping variable to **Split by** box. [Section 3.15]
- **Categorical data:**
  - **One-way Frequency Table:** **Analyses > Exploration > Descriptives.** Move categorical variable to **Variables** box. In **Statistics** section, tick **Frequency tables**. [Section 2.12]
  - **Two-way Frequency Table** (Cross-tabulation): **Analyses > Frequencies > Contingency Tables > Independent Samples.** Move variables to **Rows** and **Columns**. Request percentages in **Cells** section (e.g., **Row** or **Column** percent). [Section 2.13]
  - **Working with Summarised Data:** Enter summarised data into a new spreadsheet, including a **count** variable. In **Analyses > Frequencies > Independent Samples**, move **count** variable to **Counts** field. [Section 6.10]

## 4. Descriptive Statistics (Graphical)

- **Continuous data:**
  - **Density plot:** **Analyses > Exploration > Descriptives**. Move variable to **Variables** box. In **Plots** section, tick **Density**. [Section 1.11]
  - **Box plot:** **Analyses > Exploration > Descriptives**. Move variable to **Variables** box. In **Plots** section, tick **Box plot**. [Section 1.11]
  - **Scatter plot:** **Analyses > Exploration > Scatterplot**. Choose variables for **X-Axis** and **Y-Axis**. Add a fitted line by ticking **Linear** in the **Regression Line** section. [Section 8.13]
- **Categorical data (Bar Charts):**
  - **One Categorical Variable:** **Analyses > Exploration > Descriptives**. Move variable to Variables box. In Plots section, tick **Bar plot**. [Section 2.14]
  - **Two Categorical Variables (Contingency Tables command):** [Section 2.15]
    - **Clustered (Side by side):** **Analyses > Frequencies > Contingency Tables > Independent Samples**. Choose **Plots > Bar Plot**, select **Bar Type: Side by side**. Adjust X-axis as needed (e.g., columns).
    - **Stacked:** Choose **Bar Type: Stacked**.
    - **Stacked Relative Frequency:** Choose Y-axis: **Percentages within column**.
  - **Two Categorical Variables (surveymv module):** **Modules > surveymv > Survey Plots**. Choose Variable to plot and Grouping Variable. Set Type (Stacked bar) and Frequencies (Percentages). [Section 2.15]

## 5. Probability Distributions

jamovi does not have built-in functions to calculate probabilities; external applets are recommended.

- **Binomial** probabilities [Section 2.17]: <https://homepage.stat.uiowa.edu/~mbognar/applets/bin.html>
- **Normal** probabilities [Section 3.16]: <https://homepage.stat.uiowa.edu/~mbognar/applets/normal.html>

## 6. Confidence Intervals

- **Mean (Individual data):** **Analyses > Exploration > Descriptives**. Tick **Confidence interval for Mean**. [Section 3.17]
- **Mean (Summarised data):** Install **esci** module. **esci > Means and Medians > Single Group**. Select **Analyze summary data** tab, enter **Mean, Standard deviation, Sample size**. Tick **Extra details** for Standard Error. [Section 3.18]
- **Proportions:** **Frequencies > One Sample Proportion Tests > 2 Outcomes | Binomial Test**. Define the count variable as the analysis variable (if individual data, use actual categorical variable). Set **Test value** (for hypothesis testing) and tick **Confidence intervals**. Note: jamovi uses Clopper-Pearson interval, not Wilson. [Section 6.7]

## 7. Hypothesis Testing

- **One-Sample T-test:** **Analyses > T-Tests > One Sample T-Test**. Select **dependent variable**, enter **Test value** (hypothesised mean). [Section 4.11]
- **Independent Samples T-test and Confidence interval for Mean:** **Analyses > T-Tests > Independent Samples T-Test**. Move continuous variable to **Dependent variables**, grouping variable to **Grouping Variable**. Tick **Welch's** for unequal variances. Request **Mean difference** and **Confidence interval** in **Additional Statistics**. [Section 5.10]
- **Paired T-test:** **Analyses > T-Tests > Paired Samples T-Test**. Enter **paired variables** into **Paired Variables** field. Select **Mean difference** and **Confidence interval** in **Additional Statistics**. [Section 5.12]

Feedback? Contact me at [phcm9795.biostats@unsw.edu.au](mailto:phcm9795.biostats@unsw.edu.au)